

Exercise 2: t-test

Working memory

- A group of rhesus monkeys (n=15) performs a task involving memory after having received a placebo. Their performance is graded on a scale from 0 to 100. They are then asked to perform the same task after having received a dopamine depleting agent.
- Is there an effect of treatment on the monkeys' performance?



Coffee (bean.species tab)

- You are tired and want a decent coffee but are not sure which type to have – you have the option of Robusta or Arabica beans. You find some data where they have been tasted and rated so decide to use this to base your decision on.
- Which beans should you choose for your coffee?
- Data from the CORGIS Dataset Project <https://corgis-edu.github.io/corgis/csv/coffee/>



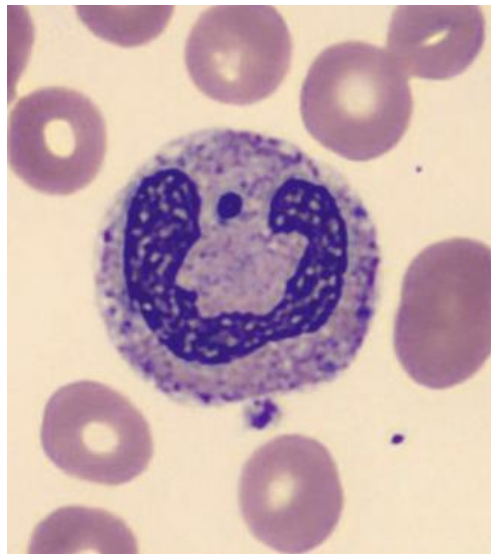
Exercise 3: ANOVA

Crop yield

- A crop researcher wants to test the effect of three different fertiliser mixtures on crop yield. They carry out an experiment using three fertiliser types (1, 2, and 3) and measure crop yield.
- Do the fertilisers have an impact on yield and, if so, which fertiliser mixture gives the highest yield?
- *From Bevens, R. (2023, June 22). One-way ANOVA | When and How to Use It (With Examples). Scribbr. <https://www.scribbr.com/statistics/one-way-anova/>*

Neutrophils

- A researcher is looking at the difference between 4 cell groups. They have run the experiment 5 times. Within each experiment, they have neutrophils from a WT (control), a KO, a KO + Treatment 1 and a KO + Treatment2.
- Is there a difference between KO with/without treatment and WT?



Exercise 4: Correlation

Roe deer

- You want to know if there is a relationship between parasite burden (PL) and body mass (BM) in roe deer. Explore the data and test this relationship.



Exam anxiety

- Is there a relationship between time spent revising and exam anxiety? And, if yes, are males and females different?
- How good is the model?

Exercise 5: Non-parametric

Oxbridge rivalry: T-shirts [Mann-Whitney]

- A group of Cambridge students hypothesise that group body odour is less disgusting when associated with an in-group member (Cambridge) versus an out-group member (Oxford). To test this, they designed a study where two groups of Cambridge University students are presented with one of two worn T-shirts with university logos. They are asked to score these on a scale of 1 to 7, with 7 being the most disgusting



- Can Cambridge students tell the difference between worn T-shirts from Oxford or Cambridge? Explore the data and answer the question with a non-parametric approach.
- What do you think about the design?

Botulinum [Wilcoxon paired]

- A group of 9 children with muscle spasticity (or extreme muscle tightness limiting movement) in their right upper limb underwent a course of injections with botulinum toxin to reduce spasticity levels.
- A neurologist (blinded) assessed levels of spasticity pre- and post-treatment for all 9 children using a 10-point ordinal scale. Higher ratings indicated higher levels of spasticity.
- Question: do botulinum toxin injections reduce muscle spasticity levels? Explore the data and answer the question with a non-parametric approach.

Creatine [Kruskal-Wallis]

- Creatine is a supplement popular among body builders, to test whether it has an impact on weight gain, three groups were tested: No creatine; Once a day; and Twice a day.
- Question: does the average weight gain depend on the creatine group to which people were assigned? Explore the data and answer the question with a non-parametric approach.

Violin [Friedman]

- An auction house is putting three violins, A, B, and C, up for bidding. Ten violinists are blindfolded and asked to rate the instruments and each player plays the violins in a randomly determined sequence (BCA, ACB, etc.).
- After each violin is played, the violinist (Rater) rates the instrument on a 10-point scale of overall excellence (Likert: 1=lowest, 10=highest).
- Question: which violin is the best according to the 10 violinists?
 - Explore the data and answer the question with a non-parametric approach.

Dominance [Spearman Rank]

- Six male colobus monkeys were ranked for social dominance and the eggs of *Trichirus* nematode per gram of monkey faeces were measured.
- Is social dominance associated with parasitism?



Exercise 6: Qualitative data

Cats & dogs

- We previously tested whether cats are more likely to line dance given different rewards: food or affection. Repeat this analysis for dogs & compare to the results from cats, graphically represent the results.

Cane toads

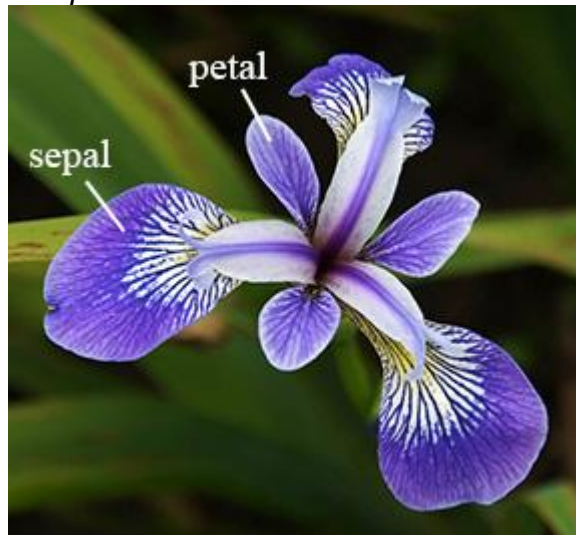
- A researcher decided to check the hypothesis that the proportion of cane toads with intestinal parasites was the same in 3 different areas of Queensland.
- Questions:
 - Is the proportion of cane toads infected by intestinal parasites the same in 3 different areas of Queensland? Produce graphical and statistical answers.
 - Is the proportion of infected cane toads different in Bowen than in the other 2 areas?
- *From Statistics Explained by Steve McKillup*



Exercise 7: Mixed

Iris flowers (flower.data)

- You want to know whether the mean petal length of iris flowers differs according to their species. You find two different species of irises growing in a garden and measure 25 petals of each species. Explore the data (species.petal tab) and test if there is a difference.
- Next, you want to know whether the mean petal length of iris flowers correlates with the mean sepal length in the virginica species. Explore the data (sepal.petal tab) and test for a correlation.
- *From Bevens, R. (2023, June 22). An Introduction to t Tests | Definitions, Formula and Examples. Scribbr. <https://www.scribbr.com/statistics/t-test/>*



Recycling

- A city wants to encourage more of its residents to recycle their household waste. They decide to test two interventions: an educational flyer (pamphlet) or a phone call. They randomly select 300 households and randomly assign them to the flyer, phone call, or control group (no intervention). They'll use the results of their experiment to decide which intervention to use for the whole city. Six months after the intervention, the city looks at the outcomes for the 300 households.
- Which intervention should the city use to maximise recycling?
- *From Turney, S. (2023, June 22). Chi-Square Test of Independence | Formula, Guide & Examples. Scribbr. <https://www.scribbr.com/statistics/chi-square-test-of-independence/>*

Intervention	Recycles	Does not recycle
Flyer (pamphlet)	89	9
Phone call	84	8
Control	86	24

Income data

- A social researcher is interested in the relationship between income and happiness. They survey 500 people whose incomes range from 15k to 75k and ask them to rank their happiness on a scale from 1 to 10.
- Generate a linear model describing the relationship between income and happiness.
- How happy would you expect someone earning 50k to be, based on your model?
- *From Bevens, R. (2023, June 22). Simple Linear Regression | An Easy Introduction & Examples. Scribbr. <https://www.scribbr.com/statistics/simple-linear-regression/>*

Coffee: part 2 (country tab)

- You enjoyed your previous coffee so much that you decide to do some more research, this time looking at which country the beans came from. Your local coffee shop sells beans from Ethiopia, Costa Rica, China, and Indonesia – which beans do you buy?